

GURU

Guru AI - SEE 2082 (2026) Model Solutions

Science & Technology (RE-1041'BP') - Full Solutions

Section 'A' (Multiple Choice Questions)

Ans 1.

(a) (iii) Excessive hunger and thirst

(b) (iv) Asuro, Bojho, Yarsagumba (Malabar nut, Calamus, Caterpillar fungus)

(c) (i) Burning of fossil fuels

(d) (ii) 490 N (Calculated using approx $m = 78.4/1.6 = 49\text{kg}$. Earth $W = 49 \times 10 = 490\text{N}$)

(e) (ii) Pascal's law

(f) (iii) Magnetic to electrical energy

(g) (ii) $1s^2, 2s^2 2p^6, 3s^2$ (Magnesium, valency 2)

(h) (i) $\text{Be} < \text{Mg} < \text{Ca}$ (Reactivity of metals increases down the group)

(i) (i) In beakers A and B, exothermic and C endothermic reaction has occurred.

(j) (ii) Siderite (Calcination is used for carbonate ores like Siderite - FeCO_3 .)

Section 'B' (Very Short Answers)

Ans 2.

- (a) The dependent variable is the solubility (or the rate of dissolving) of the solute.
- (b) (Drawing Instruction: Draw a square wave graph alternating between 0 and 1 on the Y-axis against Time on the X-axis, representing discrete values.)
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- (c) Lack of Red Blood Cells (RBCs) or hemoglobin causes anemia.
- (d) One major reason for their decline is habitat destruction (deforestation) and illegal poaching.
- (e) Weight in water = Weight in air - Upthrust (Displaced water).
- $W_{\text{water}} = 16 \text{ N} - 5 \text{ N} = 11 \text{ N}.$
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- (f) This optical illusion (mirage) is caused by Total Internal Reflection of light.
- (g) Group 1 elements are called alkali metals because they react with water to form strongly alkaline hydroxides.
- (h) The metallic property decreases as we move from left to right across a period.
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- (i) Roasting is the process of heating a concentrated sulfide ore strongly in the presence of excess air to convert it into a metallic oxide.

Section 'C' (Short Answers)

Ans 3. (Dimensional Analysis)

Given Equation: $v^2 = u^2 + 2as^2$

Unit of LHS (v^2) = $(\text{m/s})^2 = \text{m}^2/\text{s}^2$

Unit of 2nd term on RHS ($2as^2$) = $(\text{m/s}^2) \times (\text{m})^2 = \text{m}^3/\text{s}^2$

Since the units of the terms ($\text{m}^2/\text{s}^2 \neq \text{m}^3/\text{s}^2$) are not equal, the equation is dimensionally incorrect.

Ans 4. (Digital Technology)

1. It provides access to a vast amount of educational resources, e-books, and online courses anytime.
2. It enables interactive learning through 3D simulations, educational apps, and multimedia tools.

Ans 5. (Video Editing)

1. Import the long video into a video editing software.
2. Place the video on the timeline and use the "Split" or "Cut" tool to mark the start and end points of the unwanted sections.
3. Delete the unwanted sections and join the remaining clips. Export/Save the final shorter video.

Ans 6. (Octopus Classification)

1. Octopus has a soft, unsegmented body without the spiny exoskeleton characteristic of Echinodermata.
2. It lacks the water vascular system and tube feet found in echinoderms. (It is classified under Mollusca).

Ans 7. (Pteridophytes vs Bryophytes)

1. Pteridophytes have a well-differentiated plant body into true roots, stem, and leaves, unlike bryophytes.
2. Pteridophytes possess vascular tissues (xylem and phloem) for conduction, which are absent in bryophytes.

Ans 8. (Apiculture Benefits)

1. It provides highly nutritious honey and useful beeswax which have high market value.
2. Bees act as excellent pollinators, significantly increasing the yield of crops for farmers.

Ans 9. (Sex Determination Chart)

Mother (44+XX) produces eggs containing (22+X).

Father (44+XY) produces sperms containing (22+X) or (22+Y).

(Draw a Punnett square showing cross between X and X/Y gametes)

Resulting offspring: 50% chance of Female (44+XX) and 50% chance of Male (44+XY).

Ans 10. (DNA vs RNA)

Structure: DNA is a double-stranded helical molecule containing deoxyribose sugar. RNA is a single-stranded molecule containing ribose sugar.

Function: DNA stores and transfers genetic information. RNA acts as a messenger and assists in protein synthesis.

Ans 11. (Gravitation Force)

$$m_1 = 1200 \text{ kg}, m_2 = 4000 \text{ kg}, d = 4000 \text{ km} = 4 \times 10^6 \text{ m}.$$

$$F = G (m_1 m_2) / d^2 = (6.67 \times 10^{-11} \times 1200 \times 4000) / (4 \times 10^6)^2$$

$$F = 3.2016 \times 10^{-4} / 1.6 \times 10^{13} = 2.001 \times 10^{-17} \text{ N}.$$

Ans 12. (Free Fall vs Weightlessness)

1. Free fall is the motion of a body falling solely under the influence of gravity without any air resistance. Weightlessness is the sensation of having zero weight.
2. A body in free fall always experiences weightlessness, but weightlessness can occur without free fall (e.g., in deep space where gravity is negligible).

Ans 13. (Dispersion of Light)

(Drawing Instruction: Draw a triangular prism. Show a beam of white light entering it, bending, and splitting into 7 colors (VIBGYOR) emerging from the other side onto a screen. Label White Light, Prism, Spectrum, and Colors).

Ans 14. (Universe Future Graph)

(a) According to line Z, the universe will eventually stop expanding and collapse back onto itself, leading to a "Big Crunch".

(b) According to line Y, the expansion of the universe slows down and eventually remains constant (Flat Universe).

Ans 15. (Chemical Equation)

Aluminum reacts with dilute sulfuric acid to form Aluminum sulfate and Hydrogen gas.



Ans 16. (Soap vs Detergent)

Chemical Presence: Soap is a sodium or potassium salt of long-chain fatty acids. Detergent is a sodium salt of long-chain benzene sulfonic acids.

Efficiency: Soap does not form lather well in hard water (less efficient). Detergent forms lather easily even in hard water (highly efficient).

Section 'D' (Long Answers)

Ans 17. (IVF & Breeding)

In Vitro Fertilization (IVF) Process: IVF is an artificial reproductive technique where eggs are extracted from a female's ovaries and fertilized with sperm in a laboratory dish. The resulting embryos are observed for a few days, and then one or more healthy embryos are implanted into the female's uterus to establish a pregnancy. (Add a simple block diagram showing Extraction → Fertilization in dish → Incubation → Implantation).

Differences: Inbreeding involves mating between closely related individuals of the same breed, increasing homozygosity but potentially causing inbreeding depression. Line breeding is a milder form of inbreeding involving mating between distant relatives to maintain desirable traits without severe inbreeding depression.

Ans 18. (Blood Cells Report)

(a) The patient has a very low count of Cell A ($3,000/\text{mm}^3$), which represents White Blood Cells (normal range is 4,000-11,000). The problem is Leukopenia, leading to a severely weakened immune system.

(b) Blood cell 'B' (Count 3,00,000) is Platelets.

(c) If the number of cell 'C' (RBCs) is too low, the disease Anemia occurs.

(d) If bleeding from a cut does not stop, it indicates a lack of clotting factors, meaning Cell B (Platelets) might be lesser than normal in function or number.

Ans 19. (Water Density Graph)

(a) Most liquids expand when heated and contract when cooled continuously. Water, however, exhibits anomalous expansion: it contracts when heated from 0°C to 4°C and has its maximum density at 4°C , expanding thereafter.

(b) Advantage: Due to this property, water at 4°C settles at the bottom of lakes in winter, allowing aquatic life to survive under the frozen ice layer on top.

(c) (Model Graph: Draw a graph of Volume on Y-axis vs Temp on X-axis. The curve should be U-shaped, dipping to its lowest volume point precisely at 4°C .)

(d) Water cannot be used as a thermometric liquid because its expansion is non-uniform (it expands on both heating above 4°C and cooling below 4°C), and it has a high specific heat capacity and sticks to glass.

Ans 20. (Eye Defect)

(a) The defect shown (image forming behind the retina) is Hypermetropia (Long-sightedness).

(b) Cause: It is caused by the shortening of the eyeball or the lens becoming too thin (decreased converging power).

(c) (Correction Diagram: Draw the eye with a Convex Lens placed in front. The light rays pass through the convex lens, converge slightly, then enter the eye to focus perfectly on the retina.)

(d) If the image was formed at part A (in front of the retina), the defect would be Myopia, and the person would have to use a Concave lens spectacle.

Ans 21. (Transformer)

(a) Given: $N_s/N_p = 10/1$, $V_p = 220\text{ V}$.

Using formula: $V_s/V_p = N_s/N_p \rightarrow V_s/220 = 10 \rightarrow V_s = 2200\text{ V}$. The secondary voltage is 2200 V.

(b) Since the secondary voltage is greater, it is a Step-Up Transformer. (Draw a core with more turns on the secondary coil).

Ans 22. (Preparation of Gas)

(a) (Diagram: Woulfe's bottle with marble chips, thistle funnel, and delivery tube into a gas jar collecting Carbon Dioxide).

(b) Passing CO_2 in lime water for a long time makes the solution clear again as insoluble calcium carbonate becomes soluble calcium bicarbonate.



(c) Carbon dioxide is essential for plants to perform photosynthesis and synthesize food.

Ans 23. (Hydrocarbons)

(a) A homologous series is a family of organic compounds with the same general formula, similar chemical properties, and successive members differing by a $-\text{CH}_2-$ unit.

(b) (i) Compound B is Propane.

(ii) Compound A (Ethane) displaced by $-\text{OH}$ forms Ethanol. Structure: $\text{CH}_3-\text{CH}_2-\text{OH}$.

(iii) Removing 1 hydrogen from the middle carbons of Butane (Compound C) forms Butene (an Alkene). The general molecular formula is C_nH_{2n} .